E-COMMERCE · SEMESTER PROJECT 2026–2027

Bloom & Grind

Specialty coffee and brewing gear for people who brew at home

Presentation 1 — Team Setup and Planning

Avram Eliana · Ferraglio Danut

Group CEN4.H A

TEAM · GROUP CEN4.H A

Two people, each owning features end to end

Avram Eliana

Customer experience & admin

Catalog, search, cart, order history, administration panel, responsive design

Ferraglio Danut

Accounts & order pipeline

Accounts, checkout, Stripe payments, stock, security, automated tests

Both work full-stack. Every pull request is reviewed by the other member, so each of us can explain the whole codebase at P3.

NICHE & CUSTOMER

Fresh specialty beans and the gear to brew them

Target customer

Students and young professionals, 20–35, who just bought a V60, AeroPress or French press and want beans that suit it, without wading through hundreds of products.

Catalog: 5 categories, about 35 products

- Filter coffee
- Espresso coffee
- Brewers
- Grinders & scales
- Accessories

MARKET ANALYSIS

Three real stores in our niche

Store	What works	What is missing
Baristelli.ro Brașov roastery · Shopify	Free shipping from 250 lei, coffee subscriptions, bundles, wishlist, card or cash on delivery, brewing guides	Catalog drifts into craft beer, matcha and vegan powders; browsing by category only
Uzina Coffee Bucharest roastery · WordPress	Origin and process in every coffee name; curated gear such as a V60 kit and hand grinders	Only three broad categories, no filters; shipping cost appears only at checkout
Coffeedesk.com Poland, ships EU · Shopware	Filters by brew method, origin, process, roast, tasting notes and rating; roast date and recipe on product pages	Over 1,500 coffees overwhelm a beginner; mass-market and specialty mixed together

Sources: baristelli.ro, uzinacoffee.ro, coffeedesk.com — checked 12 Oct 2026

POSITIONING

What we adopt, and what we do differently

We adopt

- Origin, process and roast date on every coffee
- A free-shipping threshold shown in the cart
- Bundles of beans, filters and a brewer

We do differently

- Browse by brew method: “I brew with a V60”
- A focused catalog: only coffee and brewing gear
- Delivery cost visible in the cart, before checkout
- “Works with your brewer” suggestions computed from data

TECHNOLOGY STACK

One TypeScript codebase, from page to payment

Layer	Choice	Why
Frontend	Next.js + Tailwind CSS	Pages and API in one project; Tailwind makes 360 px layouts quick
Backend	Next.js route handlers + Zod	Our own API layer, accepted by the spec; Zod validates every input on the server
Database	PostgreSQL + Prisma ORM	Transactions for safe stock updates; parameterised queries and migrations
Auth	Auth.js sessions + bcrypt	Server-side sessions and an admin role; passwords hashed with bcrypt
Payments	Stripe test mode + Stripe CLI	Webhooks mark orders Paid; stripe listen tests them locally
Tooling	Docker Compose, Vitest, GitHub	docker compose up starts app and database; tests run with one command

PLAN · OWNERS

Every requirement has a named owner

Eliana

A1 Product catalog
A2 Basic search
A3 Shopping cart
A8 Order history
A9 Administration panel
A11 Responsive design

Danut

A4 User accounts
A5 Checkout
A6 Sandbox payment
A7 Stock management
A10 Basic security
A12 Automated tests

Planned bonuses (to confirm at P2): B1 Filtered search and B3 Reviews — Eliana · B2 Recommendations and B6 Email — Danut · 18 points

PLAN · SCHEDULE

Week by week until P3

Week	Milestone	Eliana	Danut
3	P1 · team and plan	Slides, store research	Repo, README, Docker setup
4	Foundations	Wireframes, base layout	Prisma schema, seed of 35 products
5	Catalog and accounts	A1 catalog, A2 search	A4 accounts
6	Specification	ERD, A11 layout, spec document	Architecture, payment flow, Stripe spike
7	P2 · spec and prototype	Catalog demo	Login demo
8	Cart and admin	A3 cart, A9 products	A5 checkout
9	Payments	A9 orders and statuses	A6 webhook, A7 stock
10	Feature complete	A8 history, A11 pass	A10 security, A12 tests
11	Bonuses and buffer	B1 filters, B3 reviews	B2 suggestions, B6 email
12	P3 · demo	Rehearsal, backup video	Rehearsal, final README

RISKS

Top three risks and how we handle them

01 · PAYMENT

Webhook missed or received twice

Stripe spike in week 6. Orders turn Paid only in the webhook handler, which ignores repeated events. Tested with the Stripe CLI and a declined test card.

02 · STOCK

Two customers buy the last unit

Stock drops in a database transaction with a conditional update that fails when not enough is left. An automated test runs two purchases at once.

03 · TEAM

Schedule slips with only two people

Features finish by week 10, week 11 is buffer. Weekly sync and cross-review keep either of us able to take over any part.

NEXT

Questions?

Next stop: specification and working prototype at P2, week 7

Repository: [github.com/____/bloom-and-grind]

Avram Eliana · Ferraglio Danut